(1.) Course Title: INFPHD603 Fourier analysis and its applications (complex exam preparatory)	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 100 (credit%)	
Type of Class: lecture and **number of hours**: **2/week** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*: -	
Assessment Method (Exam / Midterm Grade / **Other**[3]): colloquium **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*: -	
Course Placement in Curriculum (Semester Number): 2	
Prerequisites (*if any*): Fourier calculus I and II.	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

Orthogonal systems: orthogonal polynomials, trigonometric, Haar-, Rademacher-, Walsh systems, martingales. Wavelets, discrete and continuous versions. Construction of wavelets, applications.

Trigonometric Fourier series and Fourier transforms, inversion formula. Norm and almost everywhere convergence, different summability methods, approximation. Classical and dyadic Hardy spaces, Hilbert transform, atomic decompositions. Interpolation theory of operators. Multi-dimensional theory. Convergence of one- and multi-dimensional WalshFourier series, summability methods.

Discrete systems. Fast Fourier transforms and applications. Periodizational operator, Poisson formula. AD conversion, sampling, Shannon theorem. Uncertainty relations. Filters, window functions, multipliers. Solution of partial differential equations with Fourier method.

2-5 Most Important Required and Recommended Literature
List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

Benedetto, J. Harmonic Analysis and Applications (Studies in Advanced Mathematics), CRC Press Boca Rota, New York, London, Tokyo, 1996.
Gasquet, C., Witomski, P. Fourier Analysis and Applications: Filtering, Numerical Computation, Wavelets (Texts in Applied Mathematics 30), Springer New York, 1999.
Grafakos, L. Classical and Modern Fourier Analysis Pearson Education, New Jersey, 2004
Schipp, F.; Wade, W. R.; Simon, P. & Pál, J.:Walsh Series: An Introduction to Dyadic Harmonic Analysis. Adam Hilger, Bristol, New York, 1990
Simon, P. Fourier-transzformáció, Támop 4.2.1/B-09/1/KMR-2010-0003 (2012), 1-135.
Weisz, F. Summability of Multi-dimensional Fourier Series and Hardy Spaces Kluwer Academic Publishers, Dordrecht, Boston, London, 2002
Weisz, F. Summability of Multi-Dimensional Trigonometric Fourier Series. Surveys in Approximation Theory, 7, 1-179, 2012

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.
- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.

[1]**Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

• Engages as an equal discussion partner with professionals in the field. • Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.
Course responsible (name, position, academic degree): **Dr. Ferenc Weisz, Professor, DSc**
Instructor(s) involved in teaching the course (if any) (name, position, academic degree): -